Lab 1: Basic statistics in Python

Reference solution

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Reference Python code is in `correction.py`*; this document provides the analytical answers and expected numerical values.*

Exercise 1 – Empirical moments

Solution

(1) For $n = 10\,000$ Gaussian draws (seed 42):

$$\widehat{\mu} \approx 2.005, \quad \widehat{\sigma}^2 \approx 3.987, \quad \widehat{S} \approx -0.013, \quad \widehat{K} \approx 2.996.$$

All four are close to their theoretical values $(2, 4, 0, 3)$, with a discrepancy of order $O(1/\sqrt{n})$ for the mean and variance.
(2) For $n = 100$, the variability of $\widehat{\mu}$ is $\sigma/\sqrt{100} = 0.2$, so a typical deviation from $\mu = 2$ is of order 0.2. The plot of $\widehat{\mu}$ vs n shows convergence toward μ at rate $1/\sqrt{n}$ (Law of Large Numbers).

Exercise 2 – Method of Moments vs MLE for the Gaussian

Solution

(1) For the Gaussian, $\widehat{\mu}_{\text{MoM}} = \widehat{\mu}_{\text{MLE}} = \bar{X}_n$ and $\widehat{\sigma}^2_{\text{MoM}} = \widehat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum(X_i - \bar{X})^2$. The two functions return identical values on the same sample.
(2) On $B = 1000$ Monte-Carlo replicates with $n = 20$, $\sigma^2 = 4$:

$$\widehat{\text{Var}}_{\text{MLE}} \text{ mean } \approx 0.95\,\sigma^2 = 3.80, \qquad S_n^2 \text{ mean } \approx \sigma^2 = 4.00.$$

The factor $\frac{n-1}{n} = 0.95$ is exactly the bias predicted by theory.

Exercise 3 – Sampling distribution of $\bar{X}_n$

Solution

(1)–(2) The empirical variance of $\bar{X}_n$ over the B replicates is close to $\sigma^2/n = 4/n$:

n	σ^2/n	empirical variance
10	0.40	≈ 0.40
30	0.133	≈ 0.13
100	0.040	≈ 0.04
1000	0.004	≈ 0.004

The four histograms agree with the theoretical $\mathcal{N}(\mu, \sigma^2/n)$ density.
(3) For Exp(1) ($\mu = \sigma^2 = 1$), the histogram of $\bar{X}_n$ is visibly skewed for $n = 10$ but well-approximated by $\mathcal{N}(1, 1/n)$ from $n \geq 30$ onwards — illustrating the CLT.

Exercise 4 – CI and t-test on Atlanta temperatures

Solution

The dataset `AvTempAtlanta.txt` contains 138 years (1879–2016) of monthly temperatures, stacked into $n = 138 \times 12 = 1656$ observations.
(1) Empirical statistics: $\bar{X}_n = 61.80$, $S_n = 13.37$. The 95% t-CI is

$$\mathrm{CI}_{0.95}(\mu) = 61.80 \pm t_{1655,0.975} \cdot \frac{13.37}{\sqrt{1656}} \approx 61.80 \pm 0.64 = [61.16,\ 62.45].$$

(2) For $H_0 : \mu = 60$ vs $H_1 : \mu \neq 60$:

$$T = \frac{61.80 - 60}{13.37/\sqrt{1656}} \approx 5.486, \qquad t_{1655,0.975} \approx 1.961.$$

$|T| \gg 1.961$, so we **reject** H_0 at the 5% level. The p-value is $\approx 4 \times 10^{-8}$. The mean monthly temperature is significantly different from 60°F.
(3) `scipy.stats.ttest_1samp(x, popmean=60)` returns the same statistic and p-value (up to floating-point precision).

Exercise 5 – Bonus: bootstrap CI

Solution

With $B = 5000$ resamples on the $n = 1656$ Atlanta observations:

$$\mathrm{CI}^{\mathrm{boot}}_{0.95}(\mu) \approx [61.15,\ 62.45],$$

essentially identical to the t-CI. Both methods agree because n is large (CLT very effective). The bootstrap would be more useful in a small-sample, non-Gaussian setting where the t-approximation is questionable.